
GROWA PLATFORM

Corporate documentation establishing platform ownership, sovereign capability, and architectural intent — with trade secrets and security logic strictly protected.

CLASSIFICATION

Intellectual Property — Public-Safe

VERSION

2.0 Corporate

DATE

September 2026

APPROACH

Black-Box Architecture

GROWA IP PROTECTION

DOCUMENT CONTROL

Classification · Intellectual Property — Public-Safe Edition

Version · 2.0 Corporate

Date · September 2026

Purpose · Formal IP deposit establishing platform ownership, capability, and design intent — without disclosing trade secrets, credentials, or security-sensitive implementation.

KNOWLEDGE MAP

A minimal corporate vault. Seven interconnected notes. Black-box architecture throughout.

#	NOTE	FOCUS
01	01-Corporate-Overview	Value proposition and sovereign positioning
02	02-Black-Box-Architecture	Platform layers as protected components
03	03-Platform-Ontology	Module registry and data domains
04	04-Intelligence-Systems	Resource analytics and Growa AI
05	05-Governance-Compliance	Regulatory, supply, and institutional workflows
06	06-Trust-Boundary	Identity, isolation, and API perimeter
07	07-IP-Register	Claims, redaction policy, glossary

DESIGN PRINCIPLES

DISCLOSURE BOUNDARY

This vault communicates **what** the platform does and **why** it is proprietary.
It never exposes **how** protected algorithms, policies, or credentials operate internally.

- Warm, institutional tone — not technical documentation
- Abstract interfaces — not production source code
- Modular cards — one idea per surface

- Growa green accents — signal protected IP, not decoration

OWNERSHIP

Growa Platform and sovereign deployments (including **Growa Qatar**) are proprietary software.

All rights reserved by authorized deploying institutions.

#growa #ip-protection #corporate #sovereign-platform

CORPORATE OVERVIEW

POSITIONING

Growa is a sovereign agricultural operations platform. **Growa Qatar** is its first national deployment — a digital command centre for food security, resource accountability, and institutional governance.

How can authorized institutions see the full agricultural picture, act together, and protect national food security — without compromising data sovereignty?

Growa answers that question.

VALUE AT A GLANCE

SPATIAL INTELLIGENCE

Every metric connects to geography. The map is the operating surface, not an accessory.

RESOURCE ACCOUNTABILITY

Water and energy intensity measured per crop, per producer, per region — making sustainability visible and actionable.

SOVEREIGN CONTROL

Each country deployment is fully isolated. Data belongs to the deploying state.

GROUNDING INTELLIGENCE

AI briefings cite live platform data. No autonomous decisions. No invented metrics.

BILINGUAL BY DESIGN

English and Arabic with full RTL layout — built for Gulf institutional use.

WHO IT SERVES

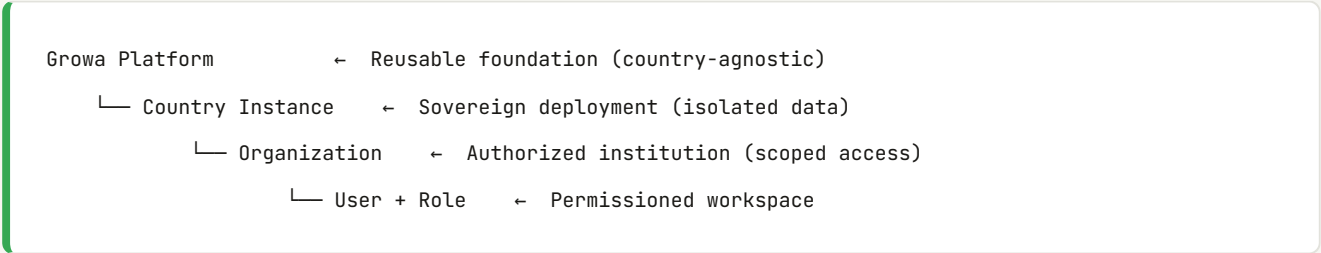
STAKEHOLDER	PRIMARY NEED
Government ministries	National oversight, policy, inspections
Inspection teams	Field cases, evidence, corrective actions
State food operators	Production and supply visibility
Financial institutions	Programme and portfolio monitoring
Farm companies	Site operations and resource tracking

Each organization sees only what it is authorized to see. Collaboration is explicit, auditable, and controlled.

PLATFORM IN ONE LINE

Growa Qatar connects maps, weather, water and energy analysis, food security planning, compliance, supply, and intelligence — so Qatar can protect food security today and plan for tomorrow.

DEPLOYMENT MODEL



See 02-Black-Box-Architecture for the protected component map.

BLACK-BOX ARCHITECTURE

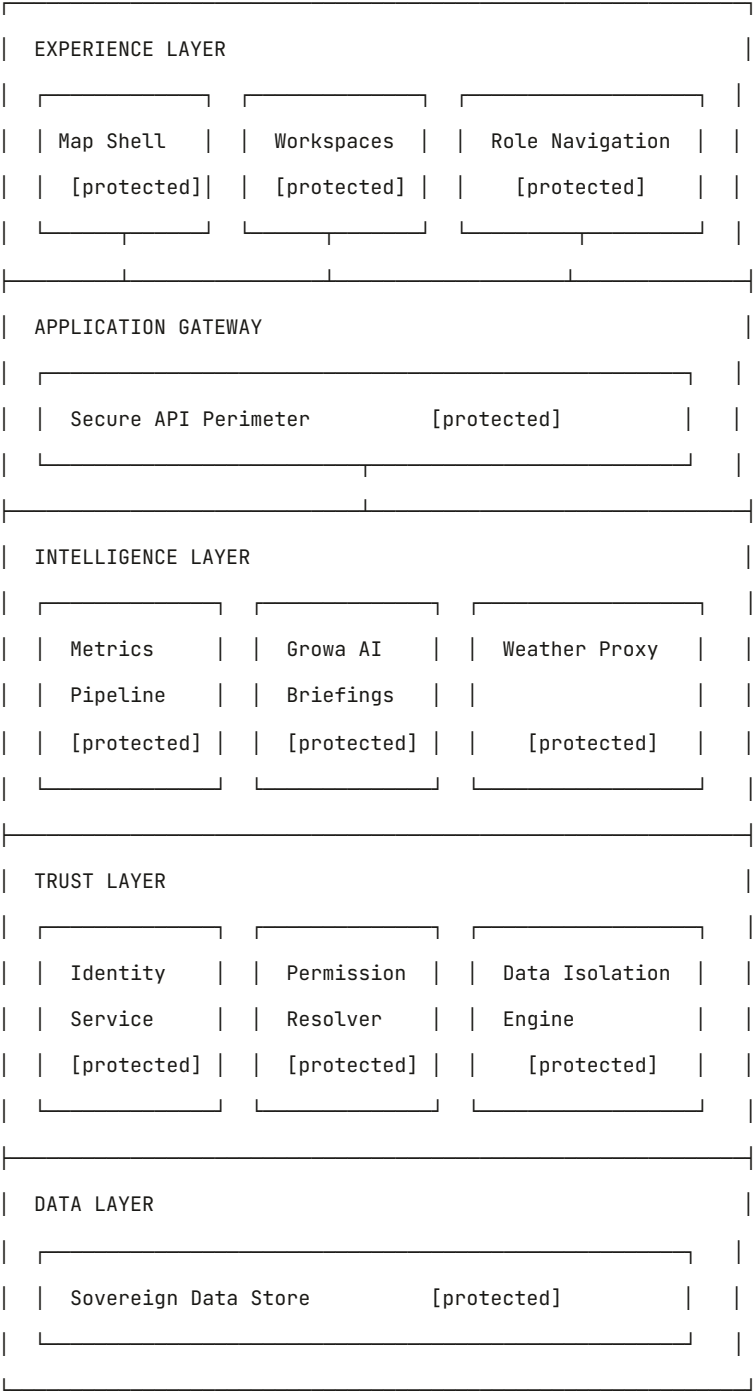
APPROACH

Growa is documented as a system of **protected components**. Each box exposes a clean interface and a clear value proposition. Internal logic, algorithms, and security predicates remain trade-secret.

BLACK-BOX RULE

Interfaces are published. Implementations are not.

SYSTEM TOPOLOGY



LAYER CONTRACTS (ABSTRACT)

EXPERIENCE LAYER

```
// Published interface - implementation protected
interface WorkspaceSurface {
  moduleId: ModuleIdentifier
  render(context: AuthorizedContext): ViewModel
  linkToMap(entities: GeoEntity[]): void
}
```

Delivers: Map-first UX, role-aware navigation, bilingual presentation.

APPLICATION GATEWAY

```
// Published interface – implementation protected
interface SecureGateway {
  authenticate(request: IncomingRequest): Session | Denied
  proxy(service: ExternalProvider, params: SafeParams): SanitizedResponse
}
```

Delivers: Server-side-only external calls. Zero credential exposure to clients.

INTELLIGENCE LAYER

```
// Published interface – implementation protected
interface IntelligencePipeline {
  aggregate(scope: NationalScope): OperationalDigest
  brief(digest: OperationalDigest, lens: AnalysisLens): ExecutiveBriefing
}
```

Delivers: Resource KPIs, producer rankings, grounded government briefings.

Algorithm internals: **protected**. See 04-Intelligence-Systems.

TRUST LAYER

```
// Published interface – implementation protected
interface PermissionResolver {
  resolve(user: Identity, resource: ResourceRef): AccessDecision
}
// AccessDecision: GRANTED | DENIED | SUMMARY_ONLY | APPROVAL_REQUIRED
```

Delivers: Six-layer permission model, database-enforced isolation.

Policy predicates: **protected**. See 06-Trust-Boundary.

COUNTRY DEPLOYMENT PATTERN

PLATFORM → COUNTRY → ORGANIZATION

One codebase. Many sovereign instances. Zero cross-country data leakage.

LAYER	CUSTOMIZABLE	SHARED
Branding & legal labels	✓	
Geography & map data	✓	
Organization presets	✓	
Core architecture		✓
Module ontology		✓
Intelligence frameworks		✓

TECHNOLOGY SURFACE (NON-SECRET)

DOMAIN	CHOICE
Application framework	Modern SSR web framework
Language	Strongly-typed application language
Data store	Relational database with row-level security
Identity	Managed authentication service
AI provider	Server-side LLM integration
UI system	Accessible component library

Specific versions and configurations: deployment-specific, not published.

PLATFORM ONTOLOGY

MODULE REGISTRY

The **Module Registry** is a core IP artifact — a declarative catalog of every workspace capability in the platform.

MODULE REGISTRY ENGINE

Defines what each module is, who sees it, and what actions it permits.
Resolution algorithm and exact visibility predicates: **protected**.

```
// Abstract module contract – not production source
interface PlatformModule {
  id:           ModuleIdentifier
  purpose:      string
  visibility:    VisibilityGate
  navigation:    RoleAwareMenu
  actions:      ActionDescriptor[]
}

interface VisibilityGate {
  orgTypes:      OrganizationClass[]
  permissions:    CapabilityFlag[]
  dataLayers:     LayerAccessMap      // implementation protected
}
```

REGISTERED CAPABILITIES

EXECUTIVE & NATIONAL

National Overview · Monitoring · Production & Harvest · Programs & Policy

SPATIAL OPERATIONS

Live Map · Farms & Sites · Custom Map Objects · Polygon Overlays

REGULATORY

Inspection Dashboard · Compliance Cases · Non-Conformities · Corrective Actions · Evidence

RISK & RESPONSE

Alerts & Risks · Inter-Agency Collaboration

ANALYTICS & REPORTING

Data Analytics · Water Intelligence · Energy Intelligence · Weather · Reports Center

INSTITUTIONAL

Supply Overview · Support · Settings

Each module ships with role-aware labels and submenus. Inspectors and directors see the same capability through different operational lenses.

NAVIGATION RESOLUTION

User signs in

- Identity resolved
- Organization + role loaded
- Visibility gates evaluated [protected]
- Authorized module surface rendered

Three resolution sources (priority order): code registry → database menus → safe fallback.

DATA DOMAINS

ACCESS & IDENTITY

Organizations · Memberships · Roles · Invitations · Audit trail

OPERATIONS

Farms · Production units · Growing cycles · Map points · Polygons · Crop insights

SUPPLY

Flow snapshots · Commodity movements · Action queue

REFERENCE

Crop catalogs · Regional taxonomies · Bilingual label sets

```
// Abstract tenancy contract – not production source
interface TenantScopedRecord {
  organizationRef: OrganizationId
  locale:          BilingualContent
  geoRef?:         GeoCoordinate
}
// All operational records inherit tenant scope.
// Enforcement mechanism: protected.
```

SHARED DATA LAYERS

Cross-cutting visibility domains applied to modules and data:

LAYER	DOMAIN
regulatory	Compliance, inspections, food security
commercial	Production, harvest, traceability
finance	Programme KPIs, dossiers
technical	Device health, diagnostics

Visibility levels: FULL · SUMMARY · APPROVAL_REQUIRED · NONE

Exact assignment rules: protected.

INTELLIGENCE SYSTEMS

PURPOSE

Growa transforms operational data into **institutional intelligence** — resource accountability, performance rankings, and executive briefings grounded in live platform metrics.

METRICS PIPELINE

Map objects → crop insights → national aggregates → workspace KPIs.
Scoring formulas and weighting logic: **protected**.

RESOURCE INTELLIGENCE

WATER INTELLIGENCE

National water consumption · Intensity (m³/t) · Irrigation pressure · Producer rankings by efficiency

ENERGY INTELLIGENCE

Sector energy use · Intensity (kWh/t) · Per-site averages · Efficiency leaders and laggards

DATA ANALYTICS

Cross-resource view · Combined efficiency scores · Crop portfolio matrix · At-risk producer flags

WEATHER GRID

National environmental grid · Agronomic risk indicators · Evapotranspiration · Disease and thermal stress signals

Every analytical table links back to the map. Numbers always have a place on the ground.

METRICS CONTRACT (ABSTRACT)

```
// Published output shape – computation protected
interface NationalSnapshot {
  scope:           NationalScope
  headline:        HeadlineKPIs
  cropMatrix:      CropPerformanceRow[]
  producerRankings: ProducerRank[]
  atRiskFlags:     RiskSignal[]
  dataGaps:        CoverageAlert[]
}

interface HeadlineKPIs {
  totalProduction: MetricValue
  totalWater:      MetricValue
  totalEnergy:     MetricValue
  averageScore:    ScoreValue
  // Module-specific extensions: protected
}
```

GROWA AI

A grounded government briefing system — not a general chatbot.

GROWA AI ENGINE

Pre-computes an operational digest before any language model invocation.
Prompt templates, scoring weights, and anti-hallucination rules: **protected**.

PIPELINE

- Live workspace metrics
- Operational digest assembled [protected]
 - Module-specific analysis lens [protected]
 - Structured briefing generated [protected]
 - Five-section executive output

OUTPUT STRUCTURE

SECTION	CONTENT
1 • Executive Summary	Key findings in plain language
2 • Evidence	Metrics cited from digest only
3 • Risk Signals	Outliers, at-risk producers, anomalies
4 • Recommended Actions	Concrete institutional interventions
5 • KPIs & Data Gaps	Targets and missing coverage

CONSTRAINTS

- Cites platform data only — never invents farms, crops, or metrics
- Server-side execution — provider keys never reach the browser
- English briefing output — human authority preserved
- Flags insufficient data explicitly

```
// Abstract briefing contract — not production source
interface GrowaBriefingRequest {
  module:    AnalysisLens
  digest:    OperationalDigest    // pre-computed, protected
  prompt?:  PresetPrompt          // template library protected
}

interface GrowaBriefingResponse {
  sections: FixedBriefingSchema  // always five sections
  sources:  DigestCitation[]     // traceable to live data
}
```

IP SIGNIFICANCE

ASSET	STATUS
Operational digest architecture	Publishable (conceptual)
Cross-resource efficiency methodology	Publishable (conceptual)
Scoring and ranking algorithms	Trade secret
AI prompt library	Trade secret
Anti-hallucination rule set	Trade secret

GOVERNANCE & COMPLIANCE

INSTITUTIONAL WORKFLOWS

Growa provides sovereign institutions with structured tools for regulatory oversight, risk management, and supply visibility.

REGULATORY SUITE

INSPECTION DASHBOARD

Inspector workload · Due items · Priority cases · Completion performance

COMPLIANCE & INSPECTIONS

Inspection planning · Non-conformity pipeline · Compliance scores · Regulatory reports

CASE MANAGEMENT

Open → Review → Evidence → Escalation → Closure lifecycle with full audit trail

NON-CONFORMITIES & CORRECTIVE ACTIONS

Severity tracking · Regional spread · Repeat patterns · Overdue remediation queues

EVIDENCE & ATTACHMENTS

Media uploads · Document library · Missing evidence alerts · Signed reports

```
// Abstract compliance contract – not production source
interface ComplianceCase {
  id:      CaseIdentifier
  state:   CaseLifecycleState
  scope:   GeoScope
  evidence: EvidenceRef[]
  audit:   ImmutableEventLog    // write mechanism protected
}
```

RISK & ALERTS

ALERTS & RISKS

Active incidents · Risk hotspots · Escalation tracking · Open vs. resolved trends

Alerts link to map locations and compliance cases. Nothing critical is lost in informal channels.

INTER-AGENCY COLLABORATION

STRUCTURED COOPERATION

Shared cases · Data requests · Approval workflows · Visibility rules · Collaboration log

```
// Abstract sharing contract – not production source
interface DataSharingRequest {
  from:      OrganizationRef
  to:        OrganizationRef
  layer:     SharedDataLayer
  visibility: VisibilityLevel      // FULL | SUMMARY | APPROVAL_REQUIRED
  status:    RequestState
}
// Approval chain logic: protected.
```

SUPPLY CHAIN VISIBILITY

SUPPLY OVERVIEW

Contract volume · Goods in transit · At-risk deliveries · Lead time trends · Sourcing action queue

Closes the loop between field production and market availability.

KPI	INSTITUTIONAL QUESTION
Contract volume	Is contracted supply sufficient?
In transit	What is moving now?
At-risk	What may miss timelines?
Lead times	Are pipelines lengthening?

FOOD SECURITY INTEGRATION

National leadership views connect production outlook, resource sustainability, compliance posture, and supply health into a single sovereign picture.

Food security is not one number. It is continuous visibility across production, resources, geography, compliance, and risk.

Workflow orchestration and cross-module data joins: **protected**.

TRUST BOUNDARY

SECURITY POSTURE

Trust is architectural, not additive. Growa enforces isolation at every layer — from the browser perimeter to the database.

TRUST BOUNDARY

Client sees authorized views only.
Server holds all secrets. Database enforces final access.
Policy logic: **protected**.

PERIMETER MODEL

CLIENT
• Session cookies (HTTP-only)
• Public configuration only
• No provider credentials
APPLICATION GATEWAY
• Session validation
• Permission pre-checks
• External service proxying
DATA ISOLATION ENGINE
• Row-level enforcement
• Membership verification
• Scope predicates [protected]

IDENTITY & ACCESS

INVITATION-ONLY ONBOARDING

No public registration. Administrators invite authorized users. Secure token activation.

SESSION MANAGEMENT

Short-lived access tokens · Automatic refresh · Invalidation on credential change

ACCOUNT STATES

Active · Pending · Suspended · Revoked

```
// Abstract access contract – not production source
interface AccessDecision {
  outcome:  'GRANTED' | 'DENIED' | 'SUMMARY' | 'PENDING_APPROVAL'
  layers:   ResolvedPermissionStack    // six layers, logic protected
  scope:    GeoAndObjectBoundary       // predicate protected
}
```

SIX-LAYER PERMISSION MODEL

- | | |
|-----------------------|----------------------------------|
| 1 · Country instance | – deployment boundary |
| 2 · Organization | – institution membership |
| 3 · Department | – subdivision scope |
| 4 · Geographic scope | – region / municipality |
| 5 · Object scope | – farm / site / unit |
| 6 · Action permission | – read / write / approve / admin |

Effective access = role template + scoped assignments. Never flattened to a single role string.

DATA ISOLATION

DEFAULT DENY

No data access unless explicitly granted by policy.

ORGANIZATION SCOPING

Every record belongs to a tenant. Cross-tenant access requires explicit sharing agreement.

DEFENSE IN DEPTH

Application checks AND database enforcement. A client bug cannot bypass isolation.

RLS policy predicates, helper functions, and RPC internals: **protected**.

API SURFACE (ABSTRACT)

CATEGORY	PURPOSE
Operations	Farm registry, map objects, crop insights
Weather	Proxied environmental data
Intelligence	Server-side AI briefings

```
// Abstract route contract – not production source
async function handleSecureRequest(
  req: AuthenticatedRequest,
  handler: AuthorizedHandler
): Promise<SafeResponse | ErrorResponse> {
  const session = await validateSession(req) // protected
  if (!session) return ErrorResponse.UNAUTHORIZED
  const allowed = await resolveAccess(session) // protected
  if (!allowed) return ErrorResponse.FORBIDDEN
  return handler(session)
}
```

ROADMAP (NON-SECRET)

CAPABILITY	PHASE
Email + password	Active
Enterprise SSO	Planned
Multi-factor authentication	Planned
Session management UI	Planned

Implementation timelines and provider configurations: deployment-specific.

IP REGISTER

PROPRIETARY ASSETS

#	ASSET	CLASSIFICATION
1	Sovereign multi-country deployment model	Publishable
2	Declarative module registry ontology	Publishable
3	Map-centric persistent shell UX	Publishable
4	Cross-resource intelligence pipeline	Concept publishable
5	Operational digest + grounded AI briefing system	Concept publishable
6	Six-layer permission resolution model	Publishable
7	Scoring, ranking, and efficiency algorithms	Trade secret
8	AI prompt and guardrail library	Trade secret
9	RLS policies and authorization RPCs	Confidential
10	Role-permission exact mappings	Trade secret

REDACTION POLICY

WHAT THIS VAULT NEVER CONTAINS

Credentials · Production URLs · Policy SQL · RPC source · Admin tooling · Live operational data · Partner identifiers · Exact permission matrices

EXCLUDED	REASON
API keys and secrets	Direct compromise risk
Infrastructure hostnames	Attack surface mapping
Security policy logic	Bypass research
Algorithm implementations	Trade secret protection
Internal QA mechanisms	Privilege escalation risk
Real organization data	Privacy and sensitivity

VERIFICATION CHECKLIST

Before any external publication or IP deposit:

- ☐ No credentials or tokens
- ☐ No production infrastructure identifiers
- ☐ No exact security predicates
- ☐ No real partner or personnel data
- ☐ Code snippets are abstract pseudocode only
- ☐ Black-box labels on all protected components

GLOSSARY

TERM	DEFINITION
Growa	Multi-country agricultural operations platform
Country Instance	Sovereign national deployment with isolated data
Module	A workspace capability in the platform catalog
Module Registry	Declarative definition of all platform modules
Black Box	Component with published interface, protected implementation
Operational Digest	Pre-computed data summary for AI briefings
Visibility Gate	Rule set determining module and data access
Shared Data Layer	Cross-cutting domain: regulatory, commercial, finance, technical
Water Intensity	m ³ of water per ton of production
Energy Intensity	kWh per ton of production
RLS	Database row-level security enforcement
BFF	Backend-for-frontend secure API perimeter

OWNERSHIP STATEMENT

Growa Platform and all sovereign deployments are proprietary software. This vault constitutes a public-safe IP deposit establishing design ownership, capability scope, and architectural intent.

